

Introductory Trigonometry

Radians, degrees and the trigonometric identities

Key topics

Distinguishing between radians and degrees

Types of triangles

Trig identities and their inverses

Sine and Cosine rules

Perimeter rule

Radians Vs Degrees

1 Radian is the angle subtended by an arc of length R, where R is the radius of the circle.

$$360^\circ = 2\pi \text{ radians} \approx 6.28 \text{ radians}$$

$$1 \text{ radian} = \frac{\text{distance travelled around circle}}{\text{radius of circle}}$$

Why bother? Degrees are an arbitrary measure (why 360 of them?) and Radians are unitless (a length divided by a length), so generally radians are more useful

Notation and conversion for angles

Degrees use the degree symbol: $^{\circ}$

Radians use c or, often, nothing at all

$$1 \text{ radian} = \frac{360^{\circ}}{2\pi} \approx 57.3^{\circ}$$

$$1^{\circ} = \frac{\pi}{180} \approx 0.0175 \text{ radians}$$

To convert from radians to degrees, multiply by $\frac{180}{\pi}$

To convert from degrees to radians, multiply by $\frac{\pi}{180}$

Warning – many sources are lazy about this notation – always question whether you are working in radians or degrees!

Right angled Triangles

The requirement that one angle be exactly 90° Allows us to make use of special relations.

Hypotenuse, h:

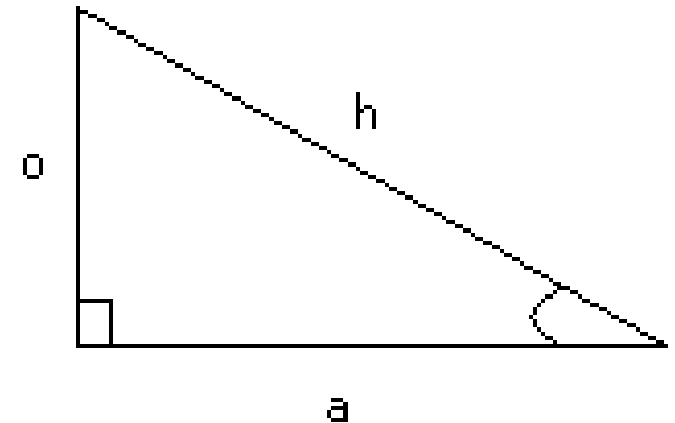
The side opposite the right angle

Adjacent, a:

The side in-between the right-angle and the acute angle (acute $< 90^\circ$)

Opposite, o:

The side opposite the acute angle

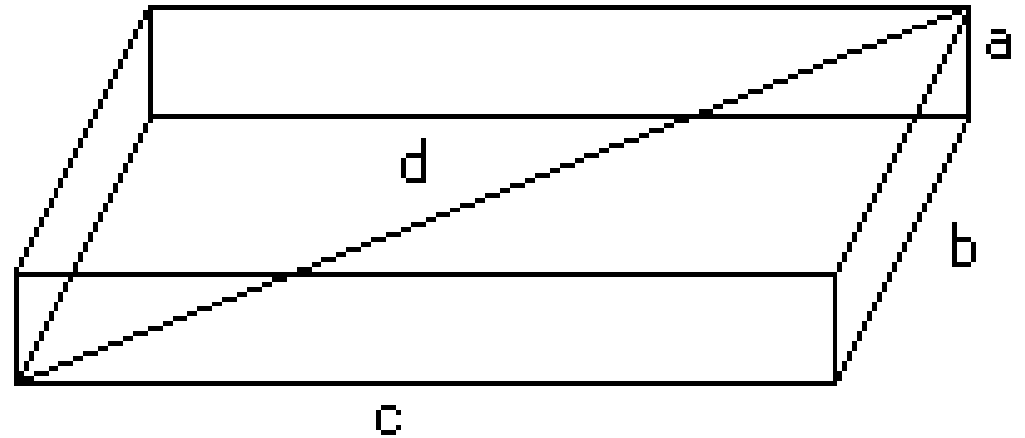


$$h^2 = o^2 + a^2$$

Extension to three dimensions

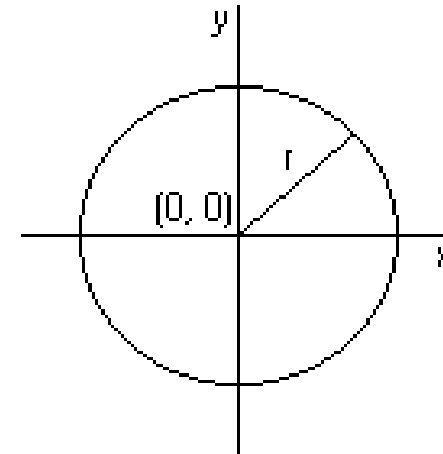
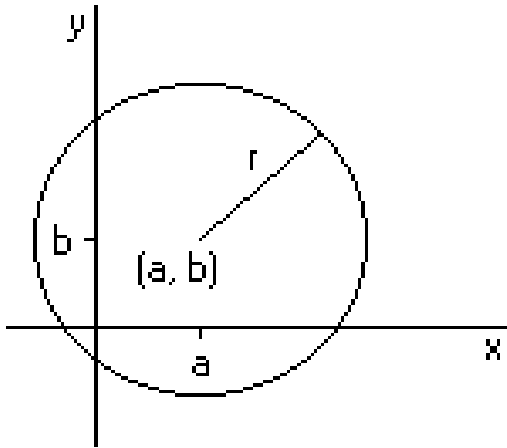
Pythagoras' equation can be scaled to higher dimensions

If we joined one corner of a cuboid to the opposite the value of d^2 would be given by $a^2 + b^2 + c^2$.



Equation of a circle

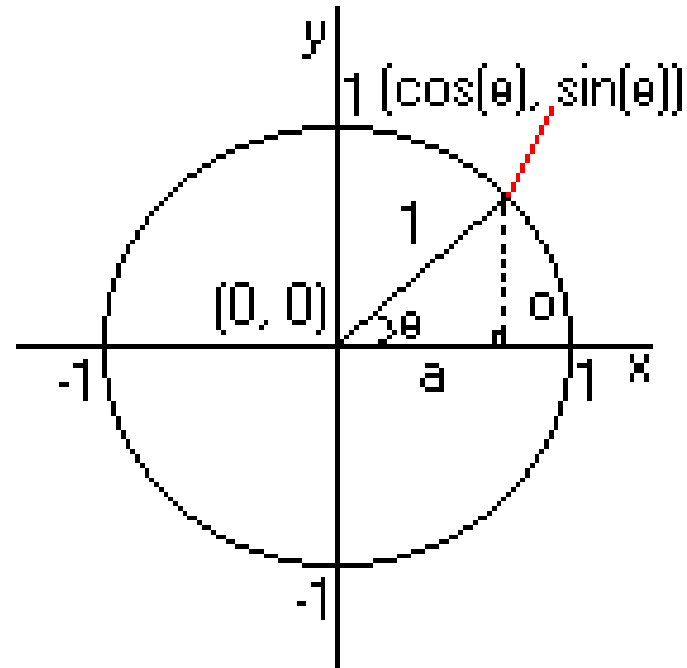
The general equation for a circle is given by, $(x - a)^2 + (y - b)^2 = r^2$, whose centre is given by (a, b) and whose radius is r .



If a circle is centred on the origin, i.e. $(0, 0)$, then $a = b = 0$. Therefore the equation of a circle centred on the origin, whose radius is r , is given by: $x^2 + y^2 = r^2$.

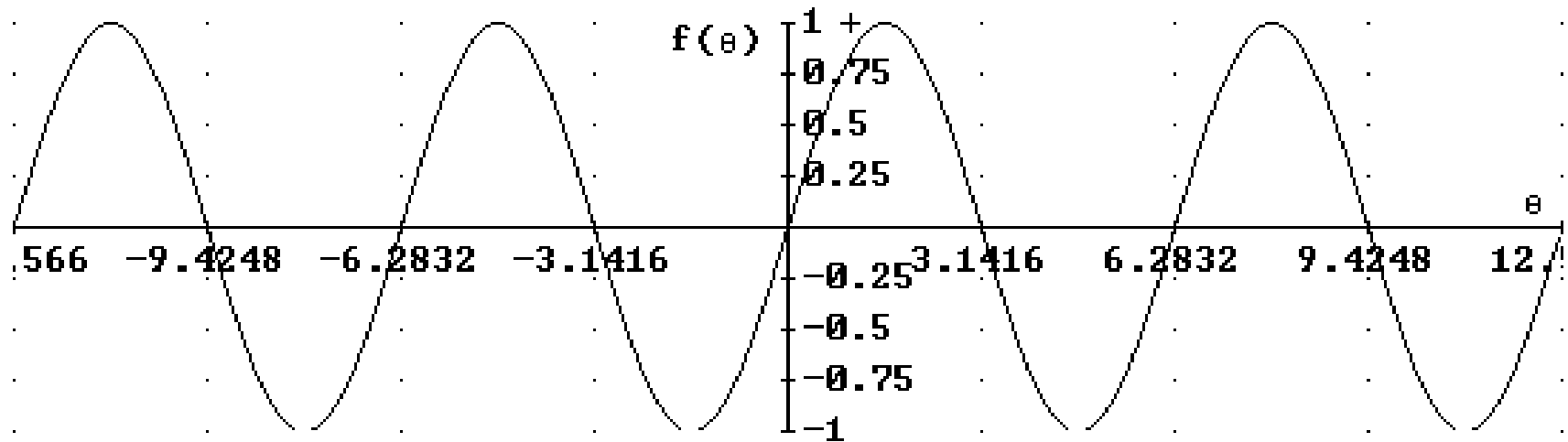
Triangles and circles

A circle centre the origin with a unit radius ($r = 1$) has the following equation: $x^2 + y^2 = 1$.



At any point on the circle's circumference the co-ordinates are $(\cos(\theta), \sin(\theta))$, where θ is measured anticlockwise from the x-axis.

Sin



Period of 2π

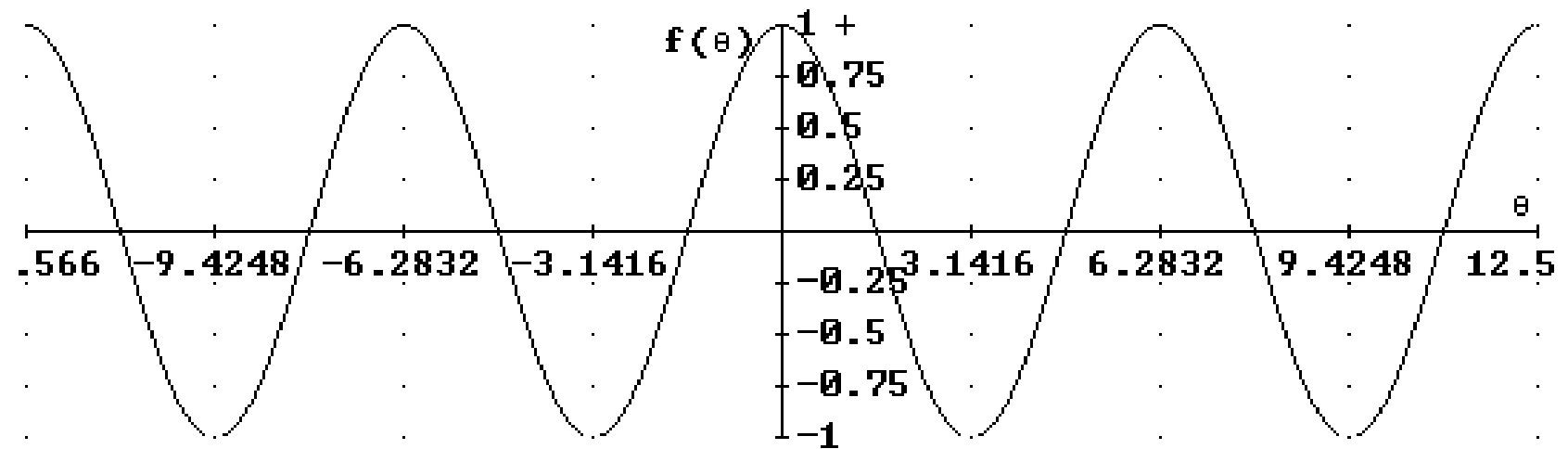
Amplitude of 1

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} \dots$$

3! Is 3 factorial, $3 \times 2 \times 1 = 6$

$$\sin(1) = 1 - \frac{1}{6} + \frac{1}{120} - \frac{1}{5040} \approx 0.841$$

Cos

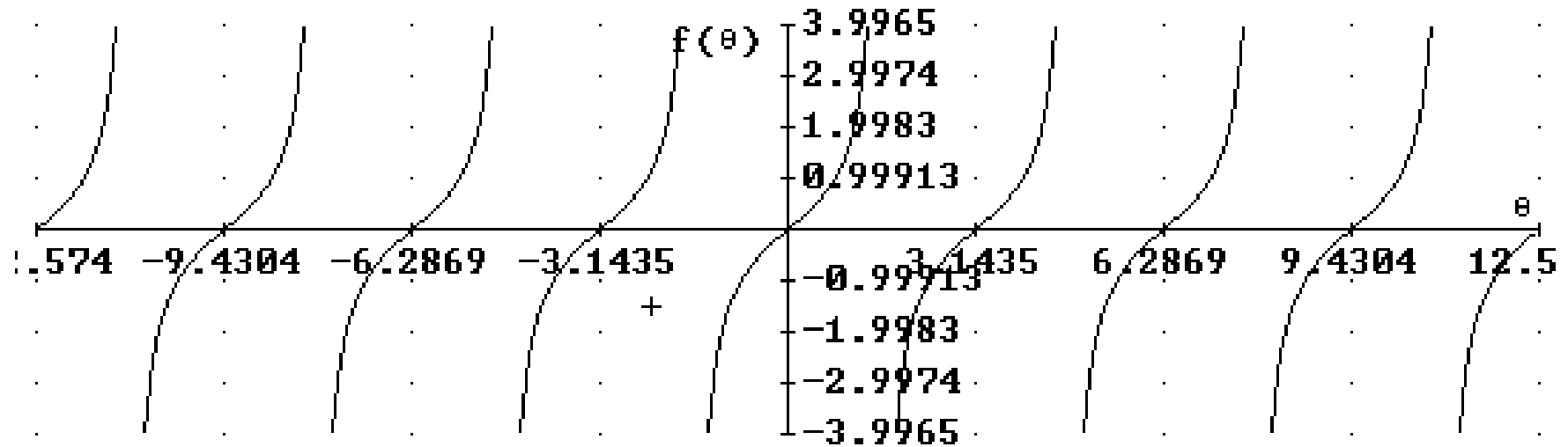


Period of 2π

Amplitude of 1

$$\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} \dots$$

Tan



Period of π

Amplitude of ∞

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\tan(x) = x + \frac{2x^3}{3!} + \frac{16x^5}{5!} + \frac{272x^7}{7!} \dots$$

SOHCAHTOA

Doesn't matter if you use degrees or radians – like using feet or inches.

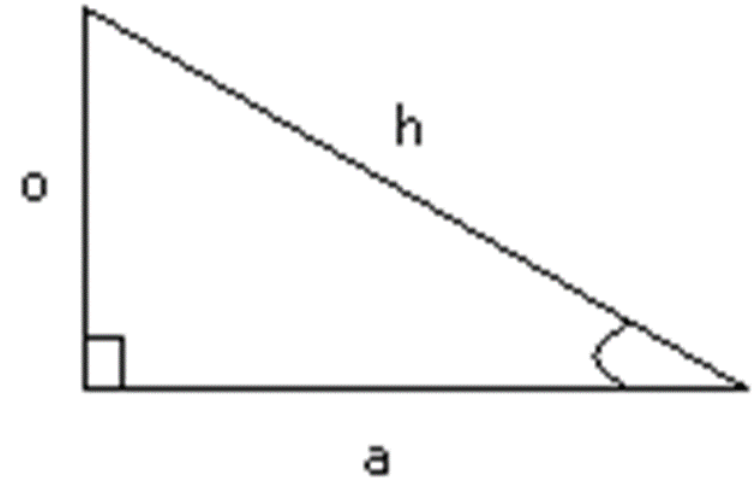
Nightmare to pronounce any acronym that uses 'H'

Better to find your own mnemonic for it:

$$\sin \theta = \frac{\textit{Opposite}}{\textit{Hypotenuse}}$$

$$\cos \theta = \frac{\textit{Adjacent}}{\textit{Hypotenuse}}$$

$$\tan \theta = \frac{\textit{Opposite}}{\textit{Adjacent}}$$



Trigonometric relations

Let $\theta=45^\circ$ We can see that $h=1$

$$\sin \theta = \frac{\text{Opposite}}{\text{Hypotenuse}}$$

$$\cos \theta = \frac{\text{Adjacent}}{\text{Hypotenuse}}$$

so $(\cos(45^\circ), \sin(45^\circ)) = (\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$.

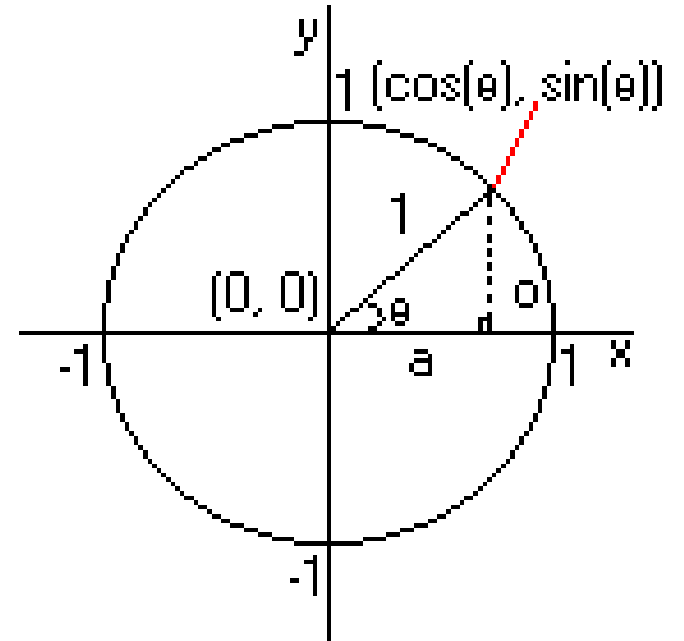
If we use the equation $x^2 + y^2 = 1$, and substitute x and y for $\cos(\theta)$ and $\sin(\theta)$ respectively, we find that

$$\cos^2 \theta + \sin^2 \theta = 1$$

e.g.

$$\cos^2 25^\circ + \sin^2 25^\circ = 1$$

$$\cos^2 55^\circ + \sin^2 55^\circ = 1$$



Inverse functions

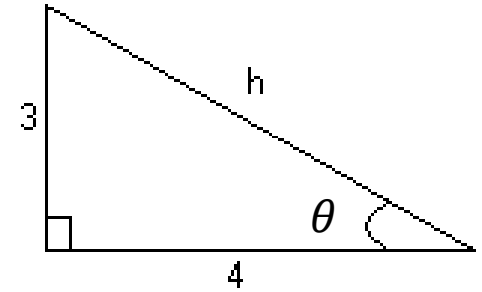
In order to use these relationships, we need to be able to transpose the formulae:

$$\sin \theta = \frac{\textit{Opposite}}{\textit{Hypotenuse}}, \quad \text{therefore} \quad \theta = \sin^{-1} \frac{\textit{opposite}}{\textit{hypotenuse}}$$

Pronounced inverse sin or arcsin. Also an arctan and arccos

Example 1

Work out θ



We know that the opposite length is 3 and the adjacent length is 4

As we know the length of the adjacent and opposite sides, we can use:

$$\tan \theta = \frac{\textit{Opposite}}{\textit{Adjacent}} = \frac{3}{4}, \quad \textit{therefore} \quad \theta = \tan^{-1} \frac{3}{4} = 36.87^\circ$$

Example 2

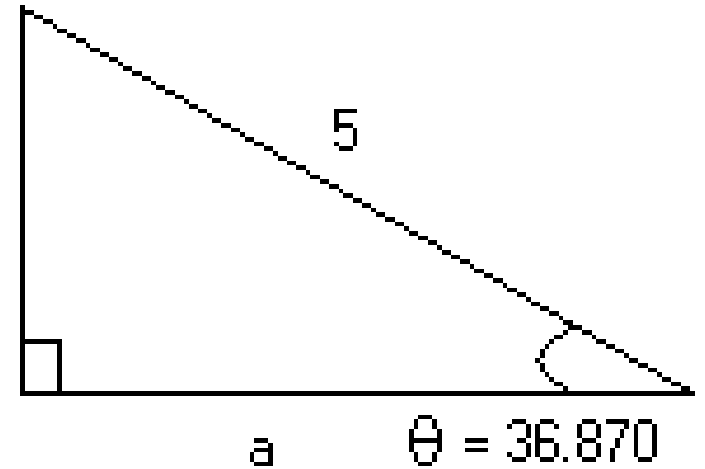
Work out a , given that $h = 5$ and $\theta = 36.870^\circ$.

We need to know a and we have h , so we use:

$$\cos \theta = \frac{\textit{Adjacent}}{\textit{Hypotenuse}}$$

$$\textit{Adjacent} = 5 \times \cos 36.870^\circ = 4$$

What do we think about the notation here?



Example 3

What are the lengths of all the triangle's sides?

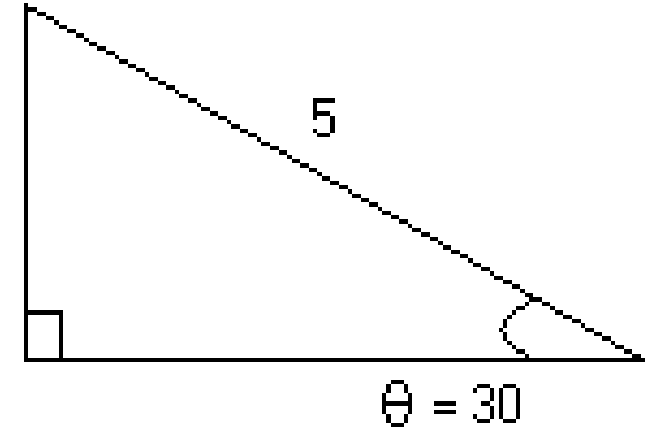
First we find what we can, knowing h and θ :

$$\sin \theta = \frac{\textit{Opposite}}{\textit{Hypotenuse}}$$

$$5 \times \sin 30^\circ = \textit{Opposite} = 2.5$$

Now we use Pythagoras' equation: $5^2 = 2.5^2 + a^2$

$$\sqrt{5^2 - 2.5^2} = a = 4.33$$

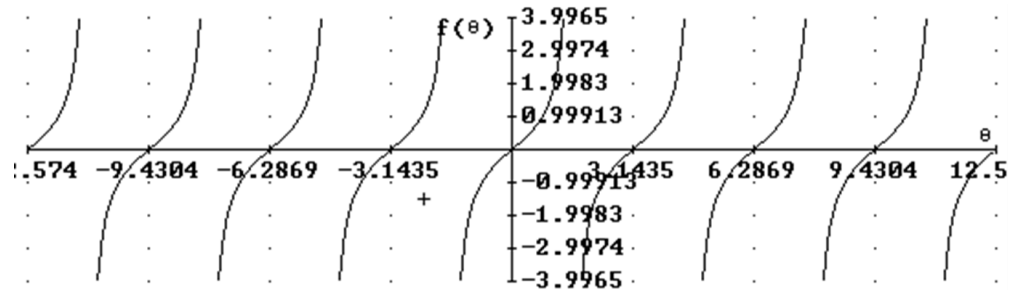
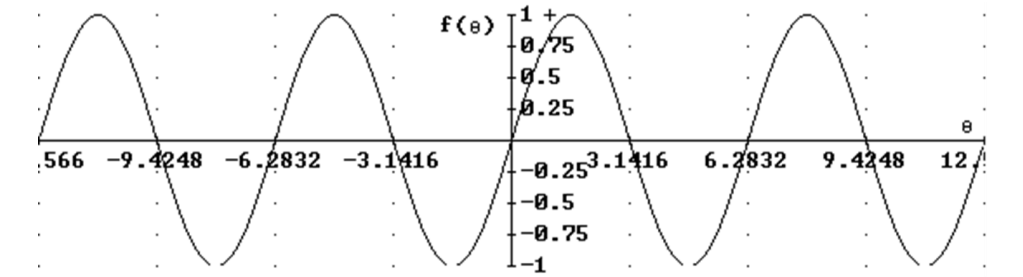
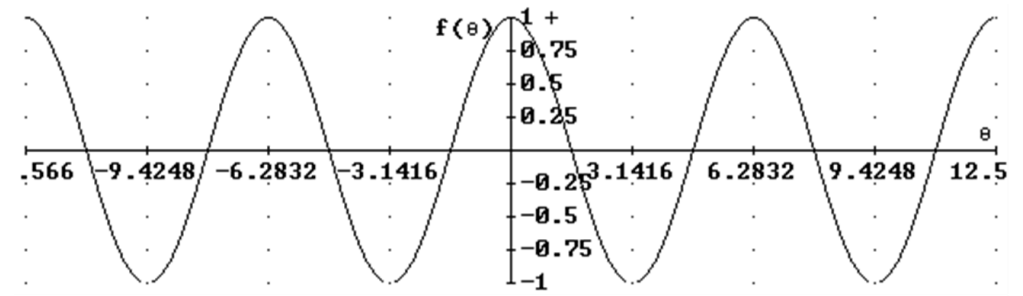


Odd and even functions

Cos is an 'even' function - it is symmetrical about the y axis

Sin and tan are odd functions

We'll return to this when we look at functions in a later class



Multiple values

It is clear that there can be an inverse to any cos value, but the function is cyclic

Consider: $\cos^{-1}(-1)$

In terms of radians, this happens at $\theta = \pi$

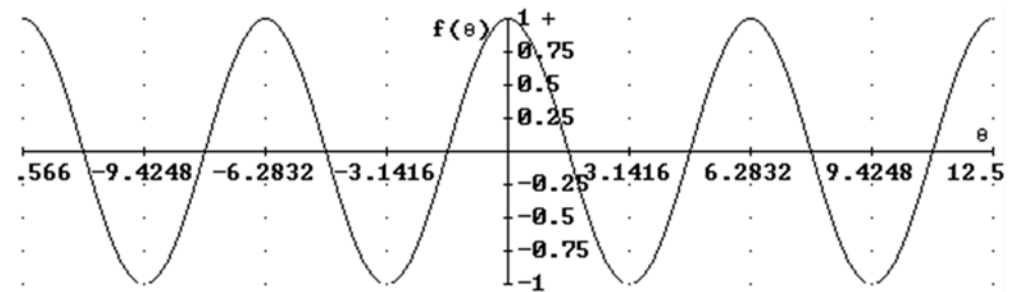
It also happens at:

$$\theta = 3\pi, \theta = -\pi, \theta = -3\pi$$

In terms of degrees, this happens at $\theta = 180^\circ$

It also happens at:

$$\theta = 540^\circ, \theta = -180^\circ, \theta = -540^\circ$$

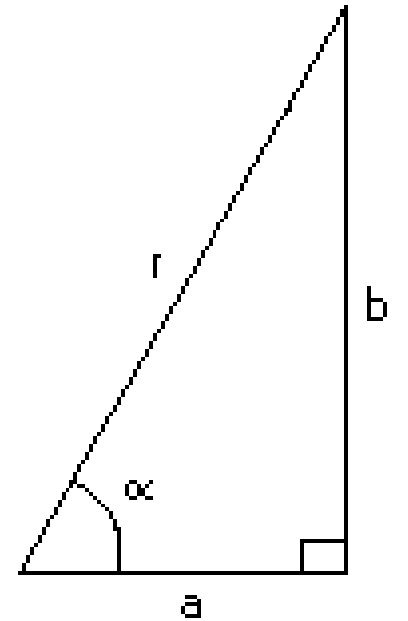


$$r \sin(\theta + \alpha)$$

Given the expression $a \sin(\theta) + b \cos(\theta)$, solve for values of θ in the range $0^\circ \leq \theta \leq 360^\circ$ (Notice that we have stated the angle in degrees)

$$r = \sqrt{a^2 + b^2}$$

$$\alpha = \tan^{-1}(b/a)$$



This is a useful format for us to view problems in, it allows us to move from cartesian coordinates to polar coordinates

Advanced Question

Solve the equation: $2 \sin \theta + 3 \cos \theta = 2$ over the range $0^\circ \leq \theta \leq 360^\circ$

If $2 \sin \theta + 3 \cos \theta = r \sin(\theta + \alpha)$ then: $r \sin(\theta + \alpha) = 2$

$$r = \sqrt{(2^2 + 3^2)} = \sqrt{13}, \alpha = \tan^{-1}\left(\frac{3}{2}\right) = 56.31^\circ$$

$$\sqrt{13} \times \sin(\theta + 56.31) = 2$$

$$\theta + 56.31 = \sin^{-1}\left(\frac{2}{\sqrt{13}}\right) = 33.69$$

Advanced Question continued

$$\theta + 56.31 = \sin^{-1}\left(\frac{2}{\sqrt{13}}\right) = 33.69$$

$$\theta = 33.69 - 56.31 = -22.62 = 337.38^\circ$$

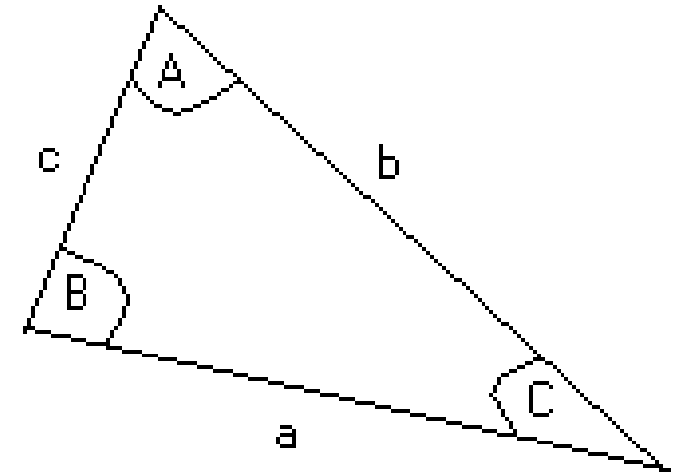
But this function is cyclic! There is another possible answer to this equation for theta in the allowed range, which also satisfies $r \sin(\theta + \alpha) = 2$ Can you find it?

Notation for non-right angled triangles

There are many choices for notation – we will adopt this one, with the angle displayed as Capital letters and the opposing lengths displayed as the lower case letter.

There are three main kinds to consider:

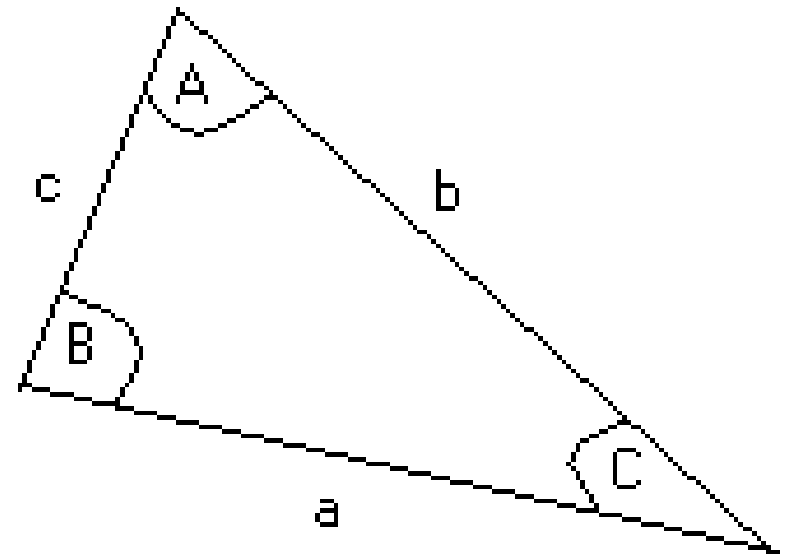
- **Scalene** - triangle with unequal sides. The word scalene is derived from the Greek word skaleños meaning unequal.
- **Equilateral** - triangle with all sides of equal length.
- **Isosceles** - triangle with two sides of equal length. The word iso is derived from the Greek word isos meaning equal. Scales is derived from skelos meaning leg - equal leg.



Sine rule

For a triangle with sides a , b , and c and angles A , B and C , we can find the ratio of any side and its opposing angle with the following sine rule:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$



Sine rule example

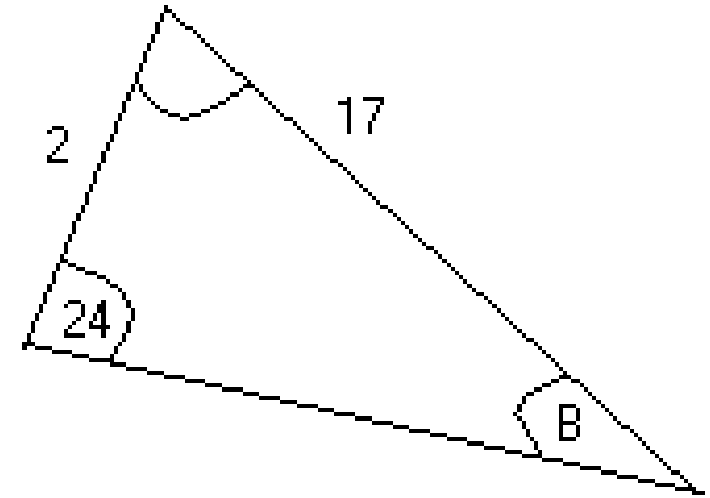
Solve the angle B , if $a = 17$, $b = 2$ and $A = 24$.

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$\frac{17}{\sin 24} = \frac{2}{\sin B}$$

$$\sin B = \frac{2}{17} \sin 24$$

$$B = \sin^{-1} \left(\frac{2}{17} \sin 24 \right) = 2.743$$



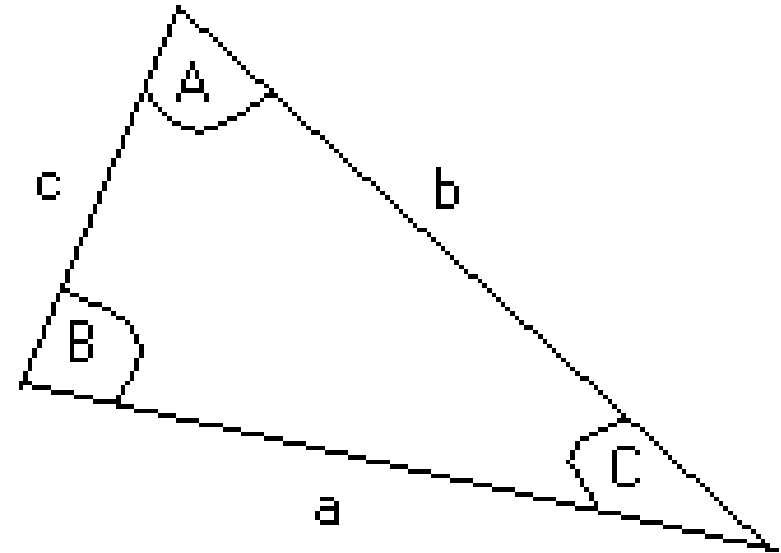
Cosine rule

For a triangle with sides a , b , and c and angles A , B and C , we can find the length of any side and the angle with the following cosine rules:

$$a^2 = b^2 + c^2 - (2bc \cos A)$$

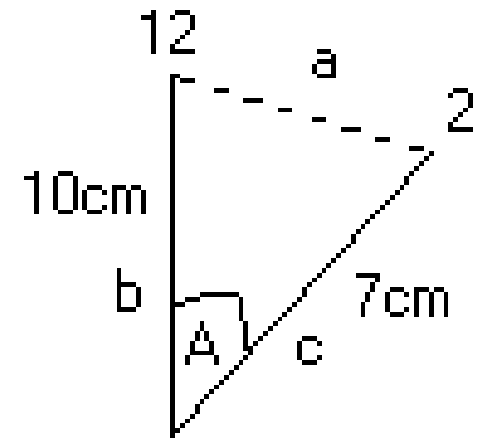
$$b^2 = a^2 + c^2 - (2ac \cos B)$$

$$c^2 = a^2 + b^2 - (2ab \cos C)$$



Cosine rule example

The hands of a clock are 10cm and 7cm in length. What is the distance between the tips of the hands when the time is 2 o'clock?



We find the angle A if we say that 12 hours is 360° , then if it is 2 o'clock the hands are $(2/12) \times 360^\circ = 60^\circ$ apart.
So $A = 60^\circ$

$$\begin{aligned}a^2 &= b^2 + c^2 - (2bc \cos A) \\a^2 &= 10^2 + 7^2 - (2 \times 10 \times 7 \cos 60) = 79 \\a &= \sqrt{79} = 8.89\text{cm}\end{aligned}$$

Answering questions

Easy questions usually require the use of one formula or relation.
Complex ones might need to do two or more operations to solve them.

First – draw a triangle, filling in all the information you know.

If you can ‘see’ what the answer needs – go for it!

If not, try to gain more information – even if it isn’t being asked for, try to fill in any missing info (look again at us finding the angle first, in the clock question).

More help?

The Foundation Maths book, by Tony Croft has a lot of examples from page 271